

Dear Harbor Satellite Maintainers and LFX Selection Committee,

I am applying for the **Harbor Satellite: Air-Gapped Peer-to-Peer Image Proxying** project. Although LFX allows applicants to pursue multiple projects, **this is the only project I applied to**. I chose it deliberately because the problem sits directly at the intersection of distributed systems, OCI infrastructure, reliability, and product-oriented systems engineering that I want to work on.

How did you find out about our mentorship program?

I found the project while going through the **LFX/CNCF mentorship project roster**. Harbor Satellite immediately stood out because the problem was concrete, technically deep, and close to systems I had already been experimenting with around peer-to-peer synchronization, OCI artifacts, content-addressed storage, digest verification, and failure recovery.

Rather than applying broadly, I decided to focus specifically on this project.

Why are you interested in this program?

What interests me is not just implementing a P2P transfer path, but understanding **when and how it becomes genuinely useful to Harbor Satellite users**.

Air-gapped factories, remote sites, vessels, isolated clusters, and bandwidth-constrained environments all create different operational requirements. A technically correct implementation is only valuable if it handles the failures and workflows operators encounter in practice.

The mentorship gives me an opportunity to work on that complete problem - from low-level distributed-system correctness to **user requirements, deployment scenarios, failure semantics, and production usefulness** - while contributing to a major open-source cloud-native project.

What experience and knowledge/skills do you have that are applicable to this program?

I have already started working on the problem before the mentorship begins.

I have:

- Drafted a detailed **RFC for trusted-peer artifact distribution in Harbor Satellite**.
- Built a **Go POC using real OCI registries** to test digest-verified transfers, root-last publication, peer failure recovery, corrupted content, and concurrent-request deduplication.
- Investigated Harbor Satellite's current replication architecture and its transition toward an ORAS-based OCI model.
- Submitted upstream work addressing **digest-correct replication** and the proposed peer-distribution architecture.
- Previously built distributed systems around **content-addressed synchronization, peer fallback, OCI image caching, concurrency, retries, and fault tolerance**.

This means I can contribute not only to implementation, but also to identifying **which requirements matter, which edge cases need to be designed for, and which additional use cases can turn the underlying mechanism into a feature operators will actually adopt.**

What do you hope to get out of this mentorship experience?

I want to become substantially better at designing and shipping **production-grade distributed infrastructure in open source.**

In particular, I want to learn from Harbor maintainers how architectural decisions are evaluated in a mature CNCF project, how distributed-system tradeoffs are handled across real deployments, and how a proposal moves from POC and RFC through review, implementation, testing, and adoption.

My goal is to leave the mentorship having contributed something that is not merely technically interesting, but **reliable, maintainable, and genuinely useful to Harbor Satellite users.**

I have already invested deeply in understanding this problem, and I would be excited to continue that work with the Harbor community.

Thank you for considering my application.

Harshita Kumari